

Student Study Performance

Individual Mini Project Overview Report • Supervised Machine Learning & Analysis

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ROLL NUMBER
70

PROJECT TYPE
Individual Mini Project

METHODOLOGY
Supervised Learning
(Python)

Executive Summary: This study builds and evaluates predictive supervised machine learning models to estimate student academic outcomes (GPA and Grade Class) using secondary data from 2,392 students. Linear Regression achieved an impressive R^2 of 0.9532, identifying school absenteeism as the single most critical determinant of academic success.

1. Problem Statement

In modern educational frameworks, identifying students at risk of academic failure before final grading cycles is critical. Traditional evaluation models act post-facto, leaving limited window for intervention.

The objective of this project is to model student performance (GPA) and categorize students into academic tiers (Grade Class) using demographic, academic habit, and extracurricular indicators. By developing an automated predictive framework and web interface, educators can proactively implement targeted tutoring and intervention programs.

2. High-Level Summary Statistics

2,392

Total Observations

15

Total Variables

0.9532

Best Regression R^2

69.73%

Classification Acc.

3. Dataset Description & Variable Mapping

The secondary dataset comprises 2,392 student records spanning 12 predictive features and 2 target variables.

VARIABLE NAME	DATA TYPE	RANGE / VALUES	DESCRIPTION & CONTEXT
GPA	Continuous Target	0.00 – 4.00	Grade Point Average (Primary Regression Target).
GradeClass	Categorical Target	0.0, 1.0, 2.0, 3.0, 4.0	Grade Class Tiers: A (≥ 3.5), B (3.0-3.5), C, D, F (< 2.0).
Absences	Discrete Integer	0 – 29 Days	Total school days missed during academic year.
StudyTimeWeekly	Continuous Float	0.00 – 19.98 Hours	Weekly dedicated study hours outside classroom.
ParentalSupport	Ordinal Categorical	0 (None) to 4 (Very High)	Level of parental academic engagement & encouragement.
Tutoring	Binary Indicator	0 = No, 1 = Yes	Participation in private or school tutoring sessions.
ParentalEducation	Ordinal Categorical	0 (None) to 4 (Higher)	Highest educational attainment level of parents.
Extracurriculars	Binary Indicators	0 = No, 1 = Yes	Includes Sports, Music, Volunteering, & Activities.

4. Data Preprocessing & Exploratory Analysis

Preprocessing Steps: Missing value audit confirmed complete data integrity (0 null values). The identifier column StudentID was removed to eliminate uninformative variance. Numerical attributes were scaled using standard Z-score normalization for linear models, and an 80/20 train-test split was executed (1,913 train / 479 test samples).

Correlation & Statistical Insights:

- Absences (r = -0.92):** Possesses an extremely strong negative linear correlation with GPA. Attendance is the paramount predictor.
- Study Time Weekly (r = +0.18):** Moderate positive correlation with GPA, indicating sustained benefit from structured study.
- Parental Support (r = +0.19) & Tutoring (r = +0.15):** Consistent positive catalysts enhancing overall student performance.

5. Model Building & Performance Evaluation

Supervised learning algorithms were evaluated across Regression (GPA Prediction) and Classification (Grade Tier Categorization) tasks.

GPA Regression Results

Model Algorithm	R ² Score	RMSE	MAE	Performance Assessment
Linear Regression	0.9532	0.1966	0.1553	Best Fit Model
Ridge Regression	0.9533	0.1966	0.1553	Robust Regularized Linear Baseline
Gradient Boosting Regressor	0.9463	0.2107	0.1646	Strong Non-Linear Ensemble
Random Forest Regressor	0.9268	0.2460	0.1908	Slightly Overfit on Training Set
Decision Tree Regressor	0.8973	0.2914	0.2322	High Variance Baseline

Grade Class Classification Results

Classification Algorithm	Accuracy	Macro F1-Score	Weighted F1-Score
Gradient Boosting Classifier	69.73%	0.52	0.69
Random Forest Classifier	69.52%	0.51	0.68
Logistic Regression	68.06%	0.49	0.67

6. Feature Importance & Key Findings

Feature importance extraction via Ensemble models demonstrates clear relative contributions across predictor variables:

- Absences (~85.4% Importance):** Each missed day of school results in an average GPA penalty of approximately 0.10 grade points.
- Weekly Study Hours (~5.8% Importance):** Every additional 5 hours of weekly study yields a ~0.15 increase in predicted GPA.
- Support Factors (~4.6% Combined):** Active parental support and private tutoring provide a cumulative GPA boost between +0.25 to +0.40 points.

7. Conclusion & Scope for Future Work

Major Findings: The project successfully demonstrates that student performance can be modeled with high statistical precision (R² > 95%). Absenteeism control is the single most actionable operational leverage point for academic institutions.

Limitations & Scope: The dataset relies on self-reported weekly study durations and discrete absence counts. Non-cognitive behavioral factors (e.g., sleep patterns, mental health, classroom engagement) were not captured in this